

XIN YAO

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RESEARCH INTERESTS

My research lies at the intersection of machine learning, wireless communication systems, and information theory. I combine theoretical analysis with algorithm and system design to develop efficient, secure, and adaptive methods for distributed learning and dynamic network control, and evaluate them on real-world wireless testbeds that I build from the physical layer up.

Keywords: reinforcement learning for wireless networks; Lyapunov optimization under real-world constraints; LoRa and low-power multi-hop networks; software-defined radio and physical-layer implementation; wireless testbeds and remote experimentation platforms; decentralized and federated learning; error feedback and communication compression; information-theoretic secure aggregation; mmWave and RIS-assisted systems; wireless sensing and indoor localization.

EDUCATION

University of Florida, Gainesville, FL, USA **08/2024 – present**

Ph.D. Candidate, Electrical and Computer Engineering

Advisor: Prof. Mingyue Ji (EdgeAI Lab)

Research: machine learning for real-world wireless systems; learning-driven network control on live testbeds; decentralized learning

University of Utah, Salt Lake City, UT, USA **08/2021 – 07/2024**

Ph.D. Studies, Electrical and Computer Engineering (transferred to the University of Florida with my advisor's research group in Aug. 2024)

Advisor: Prof. Mingyue Ji

George Washington University, Washington, DC, USA **01/2020 – 06/2021**

M.S., Electrical Engineering

Tianjin University of Science and Technology (TUST), Tianjin, China **09/2015 – 06/2019**

B.Eng., Electronic Information Engineering

RESEARCH EXPERIENCE

Doctoral Researcher, EdgeAI Lab (Advisor: Prof. Mingyue Ji) **08/2021 – present**

University of Utah (08/2021 – 07/2024) and University of Florida (08/2024 – present)

EdgeAI Lab: Real-World Wireless Communication Testbed

University of Florida · Platform for the over-the-air experiments in [C1], [C4]

- Implemented the physical-layer software stack of the lab's wireless testbed in C++: modulation and demodulation, filtering, synchronization, channel encoding and decoding, and correction of carrier-frequency offset and phase offset.
- Built Python APIs on top of the C++ core so that over-the-air experiments can be scripted, repeated, and parameterized from a single configuration; designed automated experiment workflows that adapt to heterogeneous physical-layer hardware (USRP software-defined radios and LoRa radios).

- Deployed the group’s learning algorithms (reinforcement-learning spectrum sharing, power allocation, and per-slot LoRa PHY adaptation) on both lab-scale and building-scale deployments of the testbed, and used the platform for the real-time LoRa experiments reported in [C4] and the multi-hop LoRa experiments in [C1].
- Technologies: C++, Python, USRP software-defined radios, LoRa radios, Linux.

UnionLabs: AWS-Based Remote Access and Sharing of NextG and IoT Testbeds

University of Florida · NSF-funded project (CIRC award, 2025–2027) · Poster [P2], demo [P3]

- Developed and deployed the AWS-hosted web user interface for remote access to the wireless testbeds.
- Designed the physical-layer (PHY) code for different hardware, including USRP software-defined radios and LoRa devices.
- Designed the unified *Union API* that reads and drives the different PHY implementations and connects them automatically to users’ learning or optimization algorithms, and an abstraction layer that lets users run the PHY and Union API with a single command.
- Supported collaborators from several universities in running their learning and optimization algorithms through UnionLabs, including experiments that use multiple testbeds at the same time.
- The UnionLabs and OSWireless demonstration won *Third Prize* in the Demo Competition of the Warren B. Nelms Annual IoT Conference (Dec. 2025).
- Technologies: AWS, Java, JavaScript, JSON, Python, USRP, LoRa.

Constrained Optimization for Real-World Wireless Problems with Reinforcement Learning

University of Utah and University of Florida · Publications [C4], [C5]; poster [P4]

- Derived optimization objectives with Lyapunov drift-plus-penalty (LDPP) functions under real-world constraints, turning long-term constrained problems into per-slot decisions that a learning agent can act on.
- Designed novel state representations for reinforcement-learning algorithms applied to communication-system control, and evaluated Q-learning, DDPG, and MADDPG policies both in simulation and in deployment on the EdgeAI Lab testbed.
- Proposed LDPP-MADDPG, a distributed power-allocation scheme for mmWave cellular networks that combines a Lyapunov drift-plus-penalty objective with multi-agent DDPG [C5]; an earlier MADDPG power-allocation study used measurement data from the POWDER platform [P4].
- Carried out an experimental evaluation of reinforcement learning for spectrum sharing on a real-time LoRa testbed, comparing Q-learning, DDPG, and MADDPG on live hardware [C4].
- Technologies: Python, reinforcement learning, Lyapunov optimization.

FORGE-LoRa: Network Optimization with Low-Power (LoRa) Communication

University of Florida · Publication [C1] (IEEE MILCOM 2026, accepted)

- Derived the optimization objective for multi-hop LoRa networks with Lyapunov functions under real-world constraints.
- Designed the system architecture for jointly adapting the LoRa PHY parameters on a per-slot basis.
- Building on the joint-adaptation method, designed a path discovery and optimization system that keeps multi-hop communication always live against interference and hardware outages.
- Evaluated the dynamic-control policies in simulation and in deployment on the EdgeAI Lab testbed across multiple reinforcement-learning algorithms.
- Technologies: LoRa, Python, reinforcement learning, Lyapunov optimization.

LoFiLoc: Device-Free LoRa Fingerprint Indoor Localization with Commodity Radios

University of Florida · Publication [C2] (IEEE MILCOM 2026 Workshops, accepted)

- Device-free indoor localization based on fingerprints of LoRa channel measurements collected with commodity LoRa radios, so that a target carrying no device can be localized from changes in the wireless channel.

Transformer-Enhanced DDPG for RIS-Assisted MU-MISO Systems (collaboration)

University of Florida · Publication [C3] (IEEE MILCOM 2026 Workshops, accepted)

- Contributed to a transformer-enhanced deep deterministic policy gradient (DDPG) method for the joint design of base-station beamforming and reconfigurable-intelligent-surface (RIS) phase shifts in multi-user MISO systems, with H. Zhang and O. Jiang.

A-DEFEAT: Adaptive Error Feedback for Decentralized Learning

University of Utah and University of Florida · Poster [P1]; manuscript [M1]

- Designed a new decentralized learning algorithm with an adaptive, discounted error-feedback (error-compensation) mechanism for communication compression under data heterogeneity.
- Proposed the key proof technique for the adaptive error-feedback mechanism, established convergence guarantees for the proposed algorithm, and compared its convergence rate with existing algorithms.
- Evaluated the algorithm against existing decentralized learning methods under a range of settings.
- Technologies: Python, optimization theory.

Information-Theoretic Secure Aggregation with Uncoded Groupwise Keys

University of Utah · Publications [J1], [C6] · With K. Wan, H. Sun, M. Ji, and G. Caire

- Studied the information-theoretic secure aggregation problem of federated learning, in which the server must learn only the aggregate of the users' models while tolerating user dropouts, under the practical assumption that keys are mutually independent and each key is shared, uncoded, by a group of S users.
- The work characterizes when uncoded groupwise keys are optimal: if $S > K - U$ (with K users and at most $K - U$ dropouts) a new scheme achieves the same optimal communication cost as the best coded-key scheme, whereas if $S \leq K - U$ uncoded groupwise key sharing is strictly sub-optimal.
- Implemented the proposed secure-aggregation scheme on Amazon EC2 and compared it experimentally with existing secure-aggregation schemes that use offline key sharing.

UAV-Based Wireless Communication Platform

University of Florida

- Assembled customized drones and integrated wireless communication payloads for airborne networking experiments; initialized and tuned the flight controllers; designed and mounted the customized communication payloads.
- Designed the communication protocol between ground devices and the airborne node.
- Technologies: UAV flight controllers, embedded systems, wireless payload integration.

Master's Projects

George Washington University

- **Age of Information with Energy Harvesting** (advisor: Prof. Omur Ozel). Analyzed the age of information (AoI) in M/G/1/1 and idealized infinite-buffer queueing models, including variants with packet deadlines, and investigated AoI-minimization strategies for the bufferless M/G/1/1 model with packet deadlines.
- **Secure File Transfer: Encryption and Decryption for FTP** (advisor: Prof. Tian Lan). Designed a custom encryption/decryption scheme for FTP on Linux with client-side encryption and server-side decryption of commands and data, and verified end-to-end encrypted FTP transmission.

Undergraduate Projects

- **Voice-controlled smart-home system** (advisor: Jing He). Designed STM32F103-based hardware with an LD3320 voice-recognition module; wrote firmware linking the microcontroller to speaker, PC, and LCD peripherals; tested spoken commands with system feedback.
- **License-plate recognition system** (advisor: Prof. Yang An). Built a MATLAB pipeline for image capture from surveillance cameras, filtering (Hamming window), and character segmentation and matching.
- **Android music player** (advisor: Prof. Ya-Jun Li). Developed a multi-format music player in Java using Eclipse and the Android SDK.
- **Digital counter** (advisor: Prof. Yu-liang Liu). Designed the counter circuit, selected components, and assembled, tested, and debugged the board.

PUBLICATIONS

Google Scholar: scholar.google.com/citations?user=gwTaRUkAAAAJ. My name is shown in bold.

Conference Papers

- [C1] **X. Yao**, A. Bhuyan, and M. Ji, “FORGE-LoRa: Joint per-Slot PHY Adaptation for Multi-Hop LoRa Networks,” in *Proc. IEEE Military Communications Conference (MILCOM)*, National Capital Region, USA, Oct. 2026. (Accepted, to appear.)
Joint per-slot adaptation of LoRa PHY parameters combined with path discovery and optimization so that multi-hop LoRa communication stays live under interference and hardware outages; evaluated in simulation and on the EdgeAI Lab testbed.
- [C2] **X. Yao**, A. Bhuyan, and M. Ji, “LoFiLoc: Device-Free LoRa Fingerprint Indoor Localization with Commodity Radios,” in *Proc. IEEE Military Communications Conference (MILCOM) Workshops*, National Capital Region, USA, Oct. 2026. (Accepted, to appear.)
- [C3] H. Zhang, O. Jiang, **X. Yao**, and M. Ji, “Transformer-Enhanced DDGP for Beamforming and Phase Shift in RIS-Assisted MU-MISO Systems,” in *Proc. IEEE Military Communications Conference (MILCOM) Workshops*, National Capital Region, USA, Oct. 2026. (Accepted, to appear.)
- [C4] **X. Yao**, X. Wang, H. Zhang, and M. Ji, “Experimental Evaluation of Reinforcement Learning for Spectrum Sharing on a Real-Time LoRa Testbed,” in *Proc. IEEE International Symposium on Spectrum Innovation (DySPAN)*, Washington, DC, USA, May 2026, pp. 1–8. DOI: 10.1109/DySPAN69846.2026.11571139.
Q-learning, DDGP, and MADDPG spectrum-sharing policies evaluated on live LoRa hardware in real time rather than in simulation only.
- [C5] **X. Yao**, A. Bhuyan, X. Zhang, and M. Ji, “A Novel LDPP-MADDPG Approach for Distributed Power Allocation in mmWave Cellular Networks,” in *Proc. IEEE Military Communications Conference (MILCOM), 6G and Spectrum Security for Critical Communication (6GSECC) Workshop*, Los Angeles, CA, USA, Oct. 2025.
Distributed power allocation for mmWave cellular networks that combines a Lyapunov drift-plus-penalty (LDPP) objective with multi-agent deep deterministic policy gradient (MADDPG).
- [C6] K. Wan, **X. Yao**, H. Sun, M. Ji, and G. Caire, “GroupSecAgg: Information Theoretic Secure Aggregation with Uncoded Groupwise Keys,” in *Proc. IEEE International Conference on Communications (ICC)*, Rome, Italy, May–Jun. 2023, pp. 3890–3895. DOI: 10.1109/ICC45041.2023.10279171.
Conference version of [J1].

Journal Articles

- [J1] K. Wan, **X. Yao**, H. Sun, M. Ji, and G. Caire, “On the Information Theoretic Secure Aggregation with Uncoded Groupwise Keys,” *IEEE Transactions on Information Theory*, vol. 70, no. 9, pp. 6596–6619, Sept. 2024. DOI: 10.1109/TIT.2024.3422087. Preprint: arXiv:2204.11364.

Characterizes the optimal communication cost of information-theoretic secure aggregation when keys are mutually independent and each key is shared by a group of S users: uncoded groupwise keys are optimal when $S > K - U$ and strictly sub-optimal when $S \leq K - U$. Includes an Amazon EC2 implementation compared against existing schemes with offline key sharing.

Manuscripts in Preparation or Under Review

- [M1] **X. Yao**, S. Wang, J. Perazzo, K. S. Chan, and M. Ji, “Compressed Decentralized Learning with Error-Feedback under Data Heterogeneity,” 2026.

PRESENTATIONS, POSTERS AND DEMONSTRATIONS

Conference Presentations

- [T1] “FORGE-LoRa: Joint per-Slot PHY Adaptation for Multi-Hop LoRa Networks,” IEEE Military Communications Conference (MILCOM), National Capital Region, USA, Oct. 2026. (Upcoming.)
- [T2] “Experimental Evaluation of Reinforcement Learning for Spectrum Sharing on a Real-Time LoRa Testbed,” IEEE International Symposium on Spectrum Innovation (DySPAN), Washington, DC, USA, May 2026.
- [T3] “A Novel LDPP-MADDPG Approach for Distributed Power Allocation in mmWave Cellular Networks,” IEEE MILCOM 6G and Spectrum Security for Critical Communication (6GSECC) Workshop, Los Angeles, CA, USA, Oct. 2025.
- [T4] “GroupSecAgg: Information Theoretic Secure Aggregation with Uncoded Groupwise Keys,” IEEE International Conference on Communications (ICC), Rome, Italy, Jun. 2023.
- [T5] “High-Band Frequency Spectrum for 5G” (team presentation), George Washington University, Washington, DC, USA, 2020.

Posters and Demonstrations

- [P1] “A-DEFEAT: An Adaptively Discounted Error-Feedback Algorithm for Decentralized Learning” (poster), North American School of Information Theory (NASIT), Brigham Young University, Provo, UT, USA, Jun. 2026.
- [P2] “UnionLabs: Shared Access to Heterogeneous Wireless IoT and Wireless OS Testbeds” (poster), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2025.
- [P3] “OSWireless: Automatic System to Maximize Network Throughputs” (live demonstration), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2025. *Third Prize, Demo Competition.*
- [P4] “MADDPG-Based Power Allocation Scheme Using Measurement Data from POWDER” (poster), Warren B. Nelms Annual IoT Conference, University of Florida, Gainesville, FL, USA, Dec. 2024.

TECHNICAL SKILLS

- **Programming:** Python, C/C++, Java, JavaScript, MATLAB, Arduino, JSON, XHTML/DHTML.
- **Machine learning:** reinforcement learning (Q-learning, DDPG, MADDPG, transformer-enhanced policies), federated learning, decentralized learning, communication compression and error feedback, secure aggregation, distributed and parallel computing.
- **Wireless communications and signal processing:** physical-layer implementation (modulation and demodulation, synchronization, filtering, channel encoding and decoding, carrier-frequency and phase-offset correction), software-defined radio (USRP), LoRa and low-power multi-hop networking, mmWave power allocation, RIS-assisted systems, real-world testbed development, network protocol analysis, wireless fingerprinting for localization.
- **Optimization and theory:** Lyapunov drift-plus-penalty optimization, convergence analysis of decentralized

learning, information-theoretic secure aggregation, queueing models and age of information.

- **Embedded, cloud, and systems:** STM32 microcontrollers, Android SDK (Eclipse/JDK), AWS (EC2 and web deployment), Linux, UAV flight-controller setup and payload integration.
- **Design and productivity tools:** AutoCAD, Siemens NX, Sharp3D, Adobe (Photoshop, Premiere Pro, After Effects, Acrobat), MS Office, $\LaTeX$.

HONORS AND AWARDS

- Third Prize, Demo Competition, Warren B. Nelms Annual IoT Conference, University of Florida (UnionLabs / OSWireless demonstration) 2025
- Third-Prize Scholarship, Tianjin University of Science and Technology 2017
- Advanced Individual of Social Work, Tianjin University of Science and Technology 2016
- Excellent League Member, Tianjin University of Science and Technology 2016

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